

NON-CENTRAL DISTRIBUTIONS OF THE LIKELIHOOD RATIO CRITERION FOR THE EQUALITY OF EIGENVALUES OF A SPECTRAL MATRIX*

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SUMMARY

Let $\underline{X}(t) = (X_1(t), X_2(t), \dots, X_p(t))$ be a p -dimensional zero mean stationary Gaussian time series with $\underline{\Sigma}(\omega)$ as the spectral density matrix. Let $\lambda_1(\omega), \lambda_2(\omega), \dots, \lambda_p(\omega)$ be the eigenvalues of $\underline{\Sigma}(\omega)$ with $0 < \lambda_1(\omega) < \lambda_2(\omega) \dots < \lambda_p(\omega) < \infty$. In this paper we obtain noncentral distributions of the likelihood ratio statistic for testing $H_0: \lambda_1(\omega) = \lambda_2(\omega) = \dots = \lambda_p(\omega) = \lambda(\omega)$ where $\lambda(\omega)$ is unknown for all ω , against the general alternatives.

1. Introduction

Let $\underline{X}'(t) = (X_1(t), X_2(t), \dots, X_p(t))$ be a p -dimensional zero mean, real, stationary Gaussian time series. Let $\underline{\Sigma}(\omega)$ denote the spectral matrix of $\underline{X}(t)$ so that

$$\underline{\Sigma}(\omega) = \{\sigma_{\alpha\beta}(\omega)\}, (\alpha, \beta = 1, 2, \dots, p)$$

where $\sigma_{\alpha\beta}(\omega) = \frac{1}{2\pi} \sum_{-\infty}^{\infty} R_{\alpha\beta}(s)e^{-is\omega}$ denote the cross spectral density function between $X_\alpha(t)$ and $X_\beta(t)$ and $R_{\alpha\beta}(s) = E[X_\alpha(t) \cdot X_\beta(t+s)]$

denotes the corresponding cross covariance function. We assume that for each ω the hermitian matrix $\underline{\Sigma}(\omega)$ is positive definite. Define the cross periodogram by

$$I_{\alpha\beta}(\theta_j) = \frac{1}{(2\pi T)} \left[\sum_{t=1}^T X_\alpha(t)e^{-it\theta_j} \right] \left[\sum_{s=1}^T X_\beta(s)e^{-is\theta_j} \right]^*,$$

$$\alpha, \beta = 1, 2, \dots, p;$$

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where

$$\theta_j = \frac{2\pi j}{T}, j = 0, 1, 2, \dots, \frac{T}{2}.$$

A well-known form of the estimator of $\sigma_{\alpha\beta}(\omega)$ which is consistent, is given by (see Parzen (1966))

$$\hat{\sigma}_{\alpha\beta}(\omega) = \frac{1}{n} \sum_{j=-n_1}^{n_1} I_{\alpha\beta}(\omega + \theta_j)$$

where $n = 2n_1 + 1$, n_1 being the bandwidth parameter and this estimate $\hat{\sigma}_{\alpha\beta}(\omega)$ can be alternatively written as (see Priestley, Rao and Tong (1973))

$$\hat{\sigma}_{\alpha\beta}(\omega) = \frac{1}{n} \sum_{j=1}^n I_{\alpha\beta}(\omega' + \theta_j)$$

where $\omega' = \omega - [2\pi(n_1 + 1)]/T$. The estimated spectral matrix $\underline{\hat{S}}(\omega)$ is then

$$(1.1) \quad \underline{\hat{S}}(\omega) = (\hat{\sigma}_{\alpha\beta}(\omega)) = \frac{1}{n} \underline{\hat{A}}(\omega)$$

where $\underline{\hat{A}}(\omega) = \{A_{\alpha\beta}(\omega)\}$ with $A_{\alpha\beta}(\omega) = \sum_{j=1}^n I_{\alpha\beta}(\omega' + \theta_j)$.

It is known (see Goodman (1963)) that $\underline{\hat{A}}(\omega)$ is distributed as the complex Wishart distribution $CW(\underline{\hat{A}}; p, n, \underline{\hat{S}})$ defined by

$$CW(\underline{\hat{A}}; p, n, \underline{\hat{S}}) = \{\tilde{\Gamma}_p(n)\}^{-1} |\underline{\hat{S}}(\omega)|^{-n} |\underline{\hat{A}}(\omega)|^{n-p} \exp(-\text{tr } \underline{\hat{S}}^{-1} \underline{\hat{A}})$$

where $\tilde{\Gamma}(\cdot)$ is defined in James (1964). Thus from (1.1) we see that $\underline{\hat{S}}(\omega)$ is distributed as

$CW(\underline{\hat{S}}; p, n, \frac{1}{n} \underline{\hat{S}})$. The problems of studying the structures of the above spectral matrix

and specially testing the significance of the eigenvalues of this matrix arise in the analysis of the data in numerous areas such as the vibrations of the airframe structures, meteorological forecasts and signal detection (see Hannan (1970), Liggett (1972, 1973), Priestley, Rao and Tong (1973), Brillinger (1974) and Krishnaiah (1976)). In section 2 of this paper we obtain the test of the equality of all eigenvalues of the spectral matrix $\underline{\hat{S}}(\omega)$ while in section 4 and 5 we obtain the non-null distributions of the test statistic in a form which is very well adapted on a computer so that one can compute the exact power as well as the percentage points to any degree of accuracy.

2. Test of equality of eigenvalues of $\underline{\hat{S}}(\omega)$

Since $\underline{\hat{S}}(\omega)$ is a hermitian positive definite matrix there exists a unitary matrix

$\underline{\mathbb{U}}(\omega_k)$ such that

$$\underline{\mathbb{U}} \underline{\mathbb{Z}} \underline{\mathbb{U}}^* = \underline{\mathbb{D}}$$

where for convenience we suppress ω_k and write $\underline{\mathbb{Z}}$ for $\underline{\mathbb{Z}}(\omega_k)$ etc. and $\underline{\mathbb{D}}(\omega_k)$ is a diagonal matrix with diagonal elements $\lambda_1(\omega_k), \lambda_2(\omega_k), \dots, \lambda_p(\omega_k)$ and $0 < \lambda_1(\omega_k) < \dots < \lambda_p(\omega_k)$, $\lambda_i(\omega_k)$ being the characteristic roots of $\underline{\mathbb{Z}}$.

Consider testing the hypothesis $H_0: \underline{\mathbb{Z}}(\omega_k) = \sigma(\omega_k)\mathbf{I}$ where $\mathbf{I}$ is an identity matrix and $\sigma(\omega_k)$ is scalar. The hypothesis H_0 is true if and only if all the eigen roots of $\underline{\mathbb{Z}}(\omega_k)$, which are real, are all equal. Hence testing the equality of all the eigen roots of $\underline{\mathbb{D}}$ is equivalent to testing whether $\underline{\mathbb{Z}}$ is a diagonal matrix with all its diagonal elements equal. Here we extend the "sphericity test" (see Kshirsagar (1972)).

Let $0 < g_1(\omega_k) < g_2(\omega_k) < \dots < g_p(\omega_k) < \infty$ be the eigenvalues of $\underline{\mathbb{Z}}$ defined in (1.1). These are the natural estimates of $\lambda_1(\omega_k), \dots, \lambda_p(\omega_k)$ respectively.

Now the hypothesis H_0 is a combination of the hypotheses

$H_1: \underline{\mathbb{Z}}$ is diagonal matrix

$H_2: \text{the diagonal elements of } \underline{\mathbb{Z}} \text{ are equal given that } \underline{\mathbb{Z}} \text{ is diagonal.}$

Following Anderson (1958) it can be shown that the likelihood ratio criterion for testing the hypothesis H_1 is

$$L_1 = \frac{|\underline{\mathbb{A}}|^n}{\prod_{i=1}^p |\underline{\mathbb{A}}_{ii}|^n}$$

and for testing the hypothesis H_2 we get the likelihood ratio criterion L_2 given by

$$L_2 = \frac{\prod_{i=1}^p |\underline{\mathbb{A}}_{ii}|^n}{\left[\frac{\text{Tr} \underline{\mathbb{A}}}{p} \right]^{pn}}$$

The likelihood ratio criterion L for testing H_0 is the product of L_1 and L_2 , and is given by

$$L = \frac{|\tilde{A}|^n}{\left[\frac{\text{Tr} \tilde{A}}{p} \right]^{pn}} = \left[\frac{\prod_{i=1}^p g_i \frac{1}{p}}{\sum \frac{g_i}{p}} \right]^{pn}$$

Let

$$(2.1) \quad W = (L)^{\frac{1}{n}} = \frac{|\tilde{A}|}{\left[\frac{\text{Tr} \tilde{A}}{p} \right]^p}$$

In the following sections, we proceed to obtain the non-null moments of W and use these to obtain the series representations of its non-null distribution.

3. Results required

The following results and definitions are needed in the subsequent sections. The Mellin integral transform of a function $f(x)$ of a real variable x , defined only for $x > 0$ is

$$M \{f(x)|s\} = E(x^{s-1}) = \int_0^\infty x^{s-1} f(x) dx$$

where s is any complex variate. Under suitable restrictions (Titchmarsh (1948)) satisfied by all density functions considered in this paper, there is an inverse formula or inverse Mellin transform

$$f(x) = (2\pi i)^{-1} \int_{c-i\infty}^{c+i\infty} x^{-s} M \{f(x)|s\} ds$$

valid almost everywhere. A path of integration is any line parallel to the imaginary axis and lying within the strip of analyticity of $M \{f(x)|s\}$.

Lemma 1

Let

$$\phi(t) = \int x^t p(x) dx$$

be the moment function of a random variable x with the distribution law $p(x)$. If

$$(3.1) \quad \phi(t) = O(t^{-k})$$

with real part of t tending to ∞ , then $\phi(t)$ can be expanded as a factorial series of the form

$$\phi(t) = \sum_{n=0}^{\infty} R_n \{ \Gamma(t+a) / \Gamma(t+k+n+a) \}$$

a being any arbitrary non-negative constant

(see Nair (1940), Norlund (1914)).

Lemma 2

Let the asymptotic series $\sum_{j=1}^{\infty} a_j x^j$ converge to the function $g(x)$ in the neighbourhood of $x = 0$ (or be its asymptotic expansion when $x = 0$). we then have

$$e^{g(x)} = 1 + \sum_{j=1}^{\infty} \beta_j x^j$$

where the coefficients β_j satisfy the recurrence relation

$$(3.2) \quad \beta_j = \frac{1}{j} \sum_{k=1}^j k \alpha_k \beta_{j-k}, \beta_0 = 1$$

(see Kalinin and Shalaevskii (1971)).

Lemma 3

Let $\underline{\Sigma} : m \times m$ be a Hermitian matrix i.e. $\underline{\Sigma}' = \underline{\Sigma}$.

Then

$$(3.3) \quad \tilde{F}_0(-\underline{\Sigma}^{-1}, \underline{A}) = e^{-\text{tr } \underline{A}/q} \tilde{F}_0\left(\frac{1}{q} \underline{M}, \underline{A}\right)$$

where $\underline{M} = \underline{I} - q \underline{\Sigma}^{-1}$, $\underline{A} : m \times m$ is a positive definite Hermitian matrix, q is a convergence factor ($0 < q < \infty$) and the complex hypergeometric function $\tilde{F}_0$ is as defined in equation (88) of James (1964).

Proof

We have (James (1964))

$$(3.4) \quad \tilde{F}_0(-\underline{\Sigma}^{-1}, \underline{A}) = \int_{\underline{U}(m)} e^{-\text{tr } \underline{\Sigma}^{-1} \underline{U} \underline{A} \underline{U}'} (d\underline{U})$$

where $(d\underline{U})$ is the invariant measure on the unitary group $\underline{U}(m)$ normalised to make the total measure unity. Now for any q ($0 < q < \infty$) we have

$$(3.5) \quad \text{tr } \underline{\Sigma}^{-1} \underline{U} \underline{A} \underline{U}' = \frac{1}{q} \text{tr } \underline{A} + \text{tr} \left(\frac{1}{q} \underline{M} \underline{U} \underline{A} \underline{U}' \right)$$

where $\underline{M} = \underline{I} - q \underline{\Sigma}^{-1}$. Using (3.5) in (3.4) we have (3.3).

Lemma 4

Let $\underline{A}$ be as in lemma 3 and $\underline{T} : m \times m$ be an arbitrary complex symmetric matrix.

Then

$$\widetilde{A}' \int_{\widetilde{A} > 0} \exp(-\text{tr } \widetilde{A} | \widetilde{A}|)^{a-m} \widetilde{C}_\kappa(\widetilde{A} \widetilde{T}) d\widetilde{A} = \widetilde{\Gamma}_m(a, \kappa) \widetilde{C}_\kappa(\widetilde{T})$$

where

$$\widetilde{\Gamma}_m(a, \kappa) = [a]_\kappa \cdot \widetilde{\Gamma}_m(a),$$

$[a]_\kappa$, $\widetilde{\Gamma}_m(a)$ and $\widetilde{C}_\kappa(\widetilde{T})$ are defined in section 8 of James (1964).

For proof see Khatri (1966).

Lemma 5

Let $\widetilde{T}$ and $\widetilde{A}$ be as in Lemma 4. Then

$$\begin{aligned} \widetilde{A}' \int_{\widetilde{A} > 0} \exp(-\text{tr } \widetilde{S}/q) |\widetilde{S}|^{a-m} (\text{tr } \widetilde{S})^j \widetilde{C}_\kappa(\widetilde{T} \widetilde{S}) d\widetilde{S} \\ = \widetilde{\Gamma}_m(a, \kappa) \Gamma(am + k + j) \widetilde{C}_\kappa(\widetilde{T}) q^{am+k+j} / \Gamma(am+k) \end{aligned}$$

which holds for all j provided $am + k \geq j$.

(see Pillai and Nagarsenker (1971)).

Lemma 6

Let $\widetilde{A}$: $p \times p$ be distributed as a complex Wishart $(\widetilde{A}; n, p, \widetilde{\Sigma})$ (see James (1964)). Then the non-null h -th moment of W defined in (2.1) is given by

$$(3.6) \quad E(W^h) = \frac{p^{ph}}{\widetilde{\Gamma}_p(n)} |q \widetilde{\Sigma}^{-1}|^n \sum_{\kappa=0}^{\infty} \sum_{\Sigma} \frac{\widetilde{C}_\kappa^{(M)}}{k!} \frac{\Gamma(np+k)}{\Gamma(np+k+ph)} \widetilde{\Gamma}_p(n+h, \kappa)$$

where $\underline{M} = \underline{I} - q \widetilde{\Sigma}^{-1}$. For $q = 1$, we get Equation (4.2) of Pillai and Nagarsenker (1971).

Proof

The distribution of the latent roots $g_1, g_2, \dots, g_p$ of $\widetilde{A}$ is (see James (1964))

$$(3.7) \quad k(p, n, \widetilde{\Sigma}) \widetilde{F}_0(-\widetilde{\Sigma}^{-1}, \underline{G}) |\underline{G}|^{n-p} \prod_{i < j} (g_i - g_j)^2 \prod_{i=1}^p dg_i$$

where

$$k(p, n, \underline{\Sigma}) = \frac{|\underline{\Sigma}|^{-n} \pi^{p(p-1)}}{\tilde{\Gamma}_p(n) \tilde{\Gamma}_p(p)}$$

and $\underline{G} = \text{diag}(g_1, g_2, \dots, g_p)$. Using lemma 3, (3.7) can be written as

$$(3.8) \quad k(p, n, \underline{\Sigma}) e^{-\text{tr } \underline{G}/q} {}_0F_0(\underline{M}/q, \underline{G}) |\underline{G}|^{n-p} \prod_{i < j} (g_i - g_j)^2 \prod_{i=1}^p dg_i$$

where $\underline{M} = \underline{I} - q \underline{\Sigma}^{-1}$.

Now multiplying (3.8) by $|\underline{G}| / [(\text{tr } \underline{G})/p]^p$, using the transformation $\underline{G} \rightarrow \underline{U} \underline{V} \underline{U}'$ where $\underline{U}$ is unitary and $\underline{V}$ is hermitian p.d. we have on integrating out $\underline{U}$ and using the results (see Khatri (1965)) that the Jacobian of transformation is

$$J(\underline{G}, \underline{U}, \underline{V}) = \prod_{i < j} (g_i - g_j)^2 h_2(\underline{U})$$

and that

$$\int_{\underline{U} \underline{U}' = \underline{I}} h_2(\underline{U}) d\underline{U} = \pi^{p(p-1)} \tilde{\Gamma}_p(p),$$

we have

$$E(W^h) = \frac{p^{ph} |\underline{\Sigma}^{-1}|^n}{\tilde{\Gamma}_p(n)} \sum_{k=0}^{\infty} \frac{\tilde{C}_k(\underline{q} \underline{M})}{\tilde{C}_k(\underline{I}) k!} \int_{\underline{V} \geq 0} \exp(-\text{tr } \underline{V}/q) |\underline{V}|^{n+h-p} (\text{tr } \underline{V})^{-ph} \tilde{C}_k(\underline{V}) d\underline{V}.$$

Using lemma 5 to the integral on the right we get (3.6).

4. Non-Null distribution of W

In section 4.1 we obtain exact representations of the distribution of W in terms of beta distributions while in section 4.2 we obtain its distribution in terms of gamma distributions. The representation as linear combinations of the gamma functions, though asymptotic is computationally very convenient and can be used to obtain accurate approximations even for comparatively small sample sizes and involves the computations of the coefficients d_i only using a stable recurrence relation.

4.1 Non-Null Distribution of W in Terms of Beta Distributions

From the non-null h-th moment of W given in (3.6), we have using inverse Mellin transform, the density of W as

$$(4.1) \quad f(\omega) = K(p, n, \underline{\Sigma}) \sum_{k=0}^{\infty} \sum_k \frac{\Gamma(np+k)}{k!} \tilde{C}_k(\underline{M}) \cdot p(\omega)$$

where

$$(4.2) \quad p(\omega) = (2\pi i)^{-1} \int_{c-i\infty}^{c+i\infty} \omega^{-h-1} [p^{ph} \frac{\prod_{i=1}^p \Gamma(n+h+k_i+1-i)}{\Gamma\{p(n+h)+k\}}] dh$$

and $K(p, n, \underline{\Sigma}) = |q \underline{\Sigma}^{-1}|^{n/2} \prod_{i=1}^p \Gamma(n-i+1)$. Putting $h + (n-\mu) = s$ where μ is an adjustable constant which can be chosen to govern the rate of convergence of the resulting series, we have

$$(4.3) \quad p(\omega) = \omega^{(n-\mu)-1} p^{-p(n-\mu)} \int_{d-i\infty}^{d+i\infty} \omega^{-s} \phi(s) ds$$

where

$$(4.4) \quad \phi(s) = p^{ps} \left[\prod_{i=1}^p \Gamma(s+\mu+k_i+1-i) / \Gamma\{k+p(s+\mu)\} \right].$$

The expansion of $\log \phi(s)$ will be based on the following expansion of the gamma function

$$(4.5) \quad \begin{aligned} \log \Gamma(x+h) &= (\frac{1}{2}) \log(2\pi) + (x+h - \frac{1}{2}) \log x - x \\ &- \sum_{r=1}^m \frac{(-1)^r B_{r+1}(h)}{r(r+1)x^r} + R_{m+1}(x), \end{aligned}$$

where $R_m(x)$ is the remainder such that $|R_m(x)| \leq \theta/|x^m|$, θ is a constant independent of x and $B_r(h)$ is the Bernoulli polynomial of degree r and order one defined by $\tau e^{h\tau}/(e^\tau - 1) = \sum_{r=0}^{\infty} (\tau^r/r!) B_r(h)$. Using (4.5), we obtain

$$\log \phi(s) = \log \{ p^{-p\mu-k} + \frac{1}{2} \cdot s^{-\nu} \cdot (2\pi)^{(p-1)/2} \} + \{ \sum_{r=1}^{\infty} (q_r/s^r) \}$$

where the coefficients q_r are given by

$$(4.6) \quad q_r = (-1)^{r-1} \left[\sum_{i=1}^p B_{r+1}(k_i + \mu + 1 - i) - p^{-r} B_{r+1}(k + p\mu) \right] / r(r+1)$$

and

$$\nu = (p^2 - 1)/2.$$

Thus

$$(4.7) \quad \phi(s) = s^{-\nu} (2\pi)^{(p-1)/2} p^{-p\mu-k+\frac{1}{2}} [1 + \sum_{r=1}^{\infty} (\ell_r/s^r)]$$

where the coefficients ℓ_r can be recursively computed using the recurrence relation (3.2) and (4.6). In other words the coefficients ℓ_r satisfy the recurrence relation

$$(4.8) \quad \ell_r = \frac{1}{r} \sum_{k=1}^r k q_k \ell_{r-k}, \ell_0 = 1.$$

Now from (4.3) and (4.7) we have

$$(4.9) \quad p(\omega) = K_1(p, \omega) (2\pi i) \int_{-i\infty}^{+i\infty} \omega^{-s} s^{-\nu} [1 + \sum_{r=1}^{\infty} (\ell_r/s^r)] ds$$

where

$$K_1(p, \omega) = \omega^{(n-\mu)-1} p^{\frac{1}{2}-k-pn} (2\pi)^{(p-1)/2}.$$

The integral on the right-hand side of (4.9) can be easily computed if ν is an integer and its value by Cauchy's theorem of residues is $2\pi i$ times the sum of the residues of

$$\omega^{-s} s^{-\nu} [1 + \sum_{r=1}^{\infty} (\ell_r/s^r)]$$

at $s = 0$. This is easily seen to be equal to

$$K_1(p, \omega) (2\pi i) \sum_{r=0}^{\infty} [(-\log \omega)^{\nu+r-1} \ell_r / \Gamma(\nu+r)], \ell_0 = 1$$

and terms from (4.9)

$$(4.10) \quad p(\omega) = K_1(p, \omega) \sum_{r=0}^{\infty} [(-\log \omega)^{\nu+r-1} \ell_r / \Gamma(\nu+r)]$$

and using (4.10) in (4.1), the probability that W is less than ω_0 is

$$(4.11) \quad P(W \leq \omega_0) = K_2(p, n, \Sigma) \sum_{k=0}^{\infty} \sum_{\kappa} \frac{\Gamma(np+k)}{k!} C_{\kappa}(M) \int_0^{\omega_0} p_1(\omega) d\omega$$

where

$$K_2(p, n, \underline{\Sigma}) = |q| \underline{\Sigma}^{-1} |n| p^{\frac{1}{2} - k - pn} (2\pi)^{(p-1)/2} / \prod_{i=1}^p \Gamma(n - \alpha + 1)$$

and

$$p_1(\omega) = \omega^{(n-\mu)-1} \cdot p(\omega).$$

For computational purposes, we let

$$(4.12) \quad I_{\nu+r-1, u}(\omega_0) = \int_0^{\omega_0} [\omega^u (-\log \omega)^{\nu+r-1} / \Gamma(\nu+r)] d\omega$$

where $u = (n-\mu)-1$. Then integrating by parts the R.H.S. of (4.12) it can be easily checked that the following useful recurrence relation holds:

$$(4.13) \quad I_{\nu+r-1, u}(\omega_0) = \{\omega_0^{u+1} (-\log \omega_0)^{\nu+r-1} / \Gamma(\nu+r) + I_{\nu+r-2, u}(\omega_0)\} / (u+1)$$

and

$$I_{0, u}(\omega_0) = \omega_0^{(u+1)} / (u+1).$$

With this notation, (4.11) can be written as

$$(4.14) \quad P(W \leq \omega_0) = K_2(p, n, \underline{\Sigma}) \sum_{k=0}^{\infty} \sum_{\kappa} \frac{\Gamma(np+k)}{k!} \tilde{C}_{\kappa}(\underline{M}) \cdot \sum_{r=0}^{\infty} \{ \ell_r I_{\nu+r-1, u}(\omega_0) \}$$

where $I_{\nu+r-1, u}(\omega_0)$ satisfies the recurrence relation (4.13). It is to be noted that (4.14) holds only if ν is an integer and is very efficient from the computational point of view for this case.

Remark

Let $q = \sigma^2$. Then under H_0 , $\underline{M} = 0$ and the Null distribution of W is given by

$$P(W \leq \omega_0) = \frac{p^{\frac{1}{2} - pn} (2\pi)^{\frac{1}{2} (p-1)}}{\prod_{i=1}^p \Gamma(n-i+1)} \sum_{r=1}^{\infty} \{ \ell_r I_{\nu+r-1, u}(\omega_0) \}$$

However if ν is not an integer, we can appeal to lemma 1 since in this case the moment function $E(W^h)$ satisfies (3.1) of the lemma. Thus, we can write

$$(4.15) \quad s^{-\nu} [1 + \sum_{r=0}^{\infty} (\ell_r / s^r)] = \sum_{i=0}^{\infty} R_i \Gamma(s+a) / \Gamma(s+\nu+a+i),$$

where $a \geq 0$ is any arbitrary constant which may be chosen suitably to govern the rate of convergence of the resulting series. The coefficients R_i are determined explicitly as follows.

Applying (4.5) to each gamma function on the right-hand side of (4.15) we have

$$\log \{ \Gamma(s+a) / \Gamma(s+\nu+a+i) \} = -(\nu+i) \log s + \sum_{j=1}^{\infty} (A_{ij} / s^j)$$

where

$$(4.16) \quad A_{ij} = (-1)^{j-1} [B_{j+1}(a) - B_{j+1}(\nu+a+i)] / j(j+1).$$

Thus we have

$$(4.17) \quad \{ \Gamma(s+a) / \Gamma(s+\nu+a+i) \} = s^{-(\nu+i)} \{ 1 + \sum_{j=1}^{\infty} (C_{ij} / s^j) \}$$

where using lemma 2 and (4.16), C_{ij} can be recursively computed. Using (4.17) on the right-hand side of (4.15), we have

$$(4.18) \quad s^{-\nu} [1 + \sum_{r=0}^{\infty} (\ell_r / s^r)] = \sum_{i=0}^{\infty} s^{-(\nu+i)} R_i \{ 1 + \sum_{j=1}^{\infty} (C_{ij} / s^j) \}.$$

Equating the coefficients of like powers of s on both sides of (4.18), it is easy to check the following explicit relations to determine the coefficients R_i :

$$(4.19) \quad \sum_{j=0}^i R_{i-j} C_{i-j,j} = \ell_i \quad (i = 1, 2, \dots)$$

where $R_0 = 1$ and $C_{i0} = 1$. Now using (4.15) in (4.9) and noting that term by term integration is valid since a factorial series is uniformly convergent in a half-plane (see Doetsch (1971)), we have

$$(4.20) \quad p(\omega) = K_1(p, \omega) \cdot \sum_{j=0}^{\infty} (R_j) (2\pi i)^{-1} \int_{d-i\infty}^{d+i\infty} \omega^{-s} \{ \Gamma(s+a) / \Gamma(s+\nu+j) \} ds$$

and using the well-known integral (Titchmarsh (1948))

$$\begin{aligned} & (2\pi i)^{-1} \int_{d-i\infty}^{d+i\infty} x^{-s} \{ \Gamma(s+a) / \Gamma(s+\nu+i) \} ds \\ &= x^a (1-x)^{\nu+i-1} / \Gamma(\nu+i) \quad 0 \leq x \leq 1, \end{aligned}$$

we have

$$(4.21) \quad p(\omega) = K_1(p, \omega) \sum_{j=0}^{\infty} (R_j) \omega^a (1-\omega)^{\nu+j-1} / \Gamma(\nu+j).$$

Using (4.21) in (4.1), we have the following non-central cdf of W in terms of incomplete beta functions:

$$(4.22) \quad P(W \leq \omega_0) = K_2(p, n, \underline{\Sigma}) \sum_{k=0}^{\infty} \sum_{\kappa} \frac{\Gamma(np+k)}{k!} \tilde{C}_{\kappa}(\underline{M}) \\ \cdot \sum_{i=0}^{\infty} [Q_i B_{\omega_0}(n-\mu+a, \nu+i)] / \Gamma(\nu+i)$$

where

$$Q_i = [B(n-\mu+a, \nu+i)] \cdot R_i$$

and

$$B_x(p, q) = [B(p, q)]^{-1} \int_0^x x^{p-1} (1-x)^{q-1} dx$$

is the incomplete beta function.

Remark

Taking $q = \sigma^2$, we have under null hypothesis $\underline{M} = 0$ and putting these values in (4.22), the null distribution of W is given by

$$(4.23) \quad P(W \leq \omega_0) = \frac{p^{\frac{1}{2}-pn} (2\pi)^{(p-1)/2}}{\prod_{i=1}^p \Gamma(n-\alpha+1)} \Gamma(np) \\ \cdot \sum_{i=0}^{\infty} [Q'_i B_{\omega_0}(n-\mu+a, \nu+i)] / \Gamma(\nu+i)$$

where Q'_i is the coefficient Q_i with $k_i = 0$.

4.2 Non-Null Distribution of W in Terms of Gamma Densities

Let $\lambda = -n\rho \log W$ where ρ is the adjustable constant which can be chosen to govern the rate of convergence of the resulting series. Let $G(t)$ be the characteristic function of λ . Then from (3.6) we have

$$G(t) = K(p, n, \underline{\Sigma}) \sum_{k=0}^{\infty} \sum_{\kappa} \frac{\Gamma(np+k)}{k!} \tilde{C}_{\kappa}(\underline{M}) \cdot \Psi(t)$$

where

$$\psi(t) = \frac{p^{-2n\rho pit} \prod_{i=1}^p \Gamma\{n(1-2\rho it)+k+1-\alpha\}}{\Gamma\{pn(1-2\rho it)+k\}}$$

and $K(p, n, \underline{\Sigma})$ is as given in (4.1). Using the expansion (4.5) for the gamma function, we have

$$\begin{aligned} \log \psi(t) = & \left(\frac{p-1}{2}\right) \log 2\pi - \nu \log \{n(1-2\rho it)\} \\ & - \left(pn+k-\frac{1}{2}\right) \log p + \sum_{r=1}^{\infty} \left[(D_r)/(1-2\rho it)^r\right] \end{aligned}$$

where ν is as in section 4 and the coefficients D_r are given by

$$(4.24) \quad D_r = (-1)^r n^{-r} \left[\left(\sum_{j=1}^p B_{r+1}(k_j+1-j) \right) - B_{r+1}(k)/p^r \right] / r(r+1)$$

and so

$$\begin{aligned} \psi(t) = & (2\pi)^{(p-1)/2} p^{-(pn+k-\frac{1}{2})} \{n(1-2\rho it)\}^{-\nu} \\ & \times \left[1 + \sum_{r=1}^{\infty} (d_r)/(1-2\rho it)^r \right] \end{aligned}$$

where d_r can be computed recursively using (4.24) and (3.2). Noting that $(1-\beta it)^{-\alpha}$ is the characteristic function of the gamma density $g_{\alpha}(\beta, x)$ where

$$g_{\alpha}(\beta, x) = [\beta^{\alpha} \Gamma(\alpha)]^{-1} x^{\alpha-1} e^{-x/\beta}, x > 0$$

By inverting the characteristic function $G(t)$ of λ , the cdf of $\lambda = -np \log W$ follows as

$$\begin{aligned} P(\lambda \geq \lambda_0) = & C(p, n, \underline{\Sigma}) \sum_{k=0}^{\infty} \sum_k \frac{\Gamma(np+k)}{k!} C_k(\underline{M}) \\ (4.25) \quad & \times \sum_{r=0}^{\infty} (d_r) (n^{-\nu}) G_{r+\nu}(2\rho, \lambda_0) \end{aligned}$$

where

$$C(p, n, \underline{\Sigma}) = (2\pi)^{(p-1)/2} \cdot p^{-(pn+k-\frac{1}{2})} \cdot K(p, n, \underline{\Sigma})$$

and

$$G_{\alpha}(\beta, x) = \int_x^{\infty} g_{\alpha}(\beta, x) dx$$

Remark

(1) Taking $q = \sigma^2$ and $\underline{M} = \underline{Q}$ in (4.25), the cdf of $\lambda = -n\rho \log W$ in the central case is given by

$$(4.26) \quad P(\lambda \geq \lambda_0) = (2\pi)^{(p-1)/2} \cdot p^{\frac{1}{2}-pn} [\Gamma(np) / \prod_{i=1}^p \Gamma(n-i+1)] \\ \cdot \sum_{r=0}^{\infty} (d'_r) n^{-\nu} G_{r+\nu}(2\rho, \lambda_0)$$

where d'_r are the coefficients d_r with $k_r = 0$.

(2) Further taking $\rho' = 1$ in (4.26) we obtain the following representation of the null cdf of λ in terms of chisquare distributions:

$$(4.27) \quad P(\lambda \geq \lambda_0) = (2\pi)^{(p-1)/2} \cdot p^{\frac{1}{2}-pn} [\Gamma(np) / \prod_{i=1}^p \Gamma(n-i+1)] \\ \times \sum_{r=0}^{\infty} (d'_r) (n)^{-\nu} \cdot P_r[\chi_f^2 \geq \lambda_0]$$

where $f = 2r + 2\nu$ and χ_f^2 is the chisquare with f d.f.

6. 'Conclusion

The representations (4.14), (4.22) and (4.25) of the LRC obtained in this paper enable one to compute the exact power of the tests using only the tables of zonal polynomials and tables of either incomplete beta or gamma functions and are in a form which are easily programmable. The introduction of the adjustable constants μ and ρ to govern the rate of convergence and the recurrence relations (4.8), (4.13) and (4.19) are well suited to computations of powers as well as percentage points for tests of significance. The percentage points of W had been computed in Nagarsenker and Das (1975), using the representations in (4.23) and (4.27). No numerical study is presented in this paper because of the lack of adequate computer time for the present.

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